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RACK MOUNTABLE PANEL FOR OPTIMIZING SLACK STORAGE AND MANAGEMENT OF OPTICAL FIBER CABLES

Abstract

A cable management panel for optimizing management and storage of a slack portion of a fiber optic cable includes a base portion and a cable support portion that is configured to extend from the base portion. The cable support includes a curved portion configured to prevent a fiber optic cable that is wrapped around the curved portion from being bent beyond a minimum bend radius. The cable support is configured to provide a path structure for optimizing management and storage of different lengths of slack of fiber optic cables such that a selected path length from an end of the base portion to the curved portion of the cable support and back to the end of the base portion substantially matches a length of slack of a fiber optic cable such that the length of slack is maintained within the height of the base so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 17/984,841, filed Nov. 10, 2022, which claims the benefit of U.S. Provisional Application No. 63/263,827, filed Nov. 10, 2021, the disclosures of which are hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The present disclosure generally relates to an optical fiber communication system and, more particularly, to a rack mountable panel for optimizing slack storage and management of optical fiber cables in an optical fiber communication system.

BACKGROUND

[0003] Conventional optical fiber distribution systems may include a rack or frame having several bays for receiving connector modules, associated jumper organizers for storing extra lengths of cables (which may also be called pigtailed or patch cords and comprise connectorized optical fibers which are connected to the connector modules), and upper and lower troughs for conveying jumper fibers between adjacent bays. Extra lengths of cables are typically stored in vertical passageways or raceways around the rack and outside of the bays to which the connector modules are mounted. The vertical raceways typically contain several structures, for example, spools, for managing slack cables.

[0004] However, such storage systems provide poor organization of the spare jumper lengths, and tangling of cables is therefore possible, and even likely. This tangling leads to the disturbance of other cables when attempting to access one particular jumper, which can cause losses in optical signal transmission. Furthermore, the tangling of cables is exacerbated by the fact that the vertical raceways (where the spare jumper lengths are stored) are also used to route cables between the bays of the distribution system, and so these raceways are normally very crowded with cables. This also makes the provision of wide vertical raceways between the bays a necessity, which is disadvantageous in situations where there is insufficient space available for such wide vertical raceways.

[0005] It may be desirable to provide a cable management panel for optimizing slack storage and management of optical fiber cables in an optical fiber communication system so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted. It may be desirable to provide such a cable management panel that is mountable in a standard rack mount system. It may be desirable to provide such a cable management panel in a 2 rack unit (2U) configuration.

SUMMARY

[0006] According to various aspects of the disclosure, a cable management panel for optimizing management and storage of a slack portion of a fiber optic cable includes a base that is configured to include a planar portion having a first end and a second end in a longitudinal direction and a curved portion extending from the first end of the planar portion, a cable support that is configured to extend substantially perpendicularly from the planar portion of the base, a cable management structure that is disposed proximate the first end of the planar portion and configured to extend substantially perpendicularly from the planar portion of the base in a same direction as the cable

support, and a cable retaining structure that is disposed along a longitudinal edge of the planar portion and configured to extend substantially perpendicularly from the planar portion of the base in the same direction as the cable support. The base has a height equal to two rack units, and the base is configured to be mounted in a rack having a standard rack length. The cable retaining structure is configured to retain a slack fiber optic cable between the longitudinal edge and an opposed longitudinal edge of the base, the curved portion of the base is configured to prevent a fiber optic cable that is wrapped around the curved portion of the base from being bent beyond a minimum bend radius, and the cable support includes a curved portion configured to prevent a fiber optic cable that is wrapped around the curved portion from being bent beyond a minimum bend radius. The cable support is configured to provide a path structure for optimizing management and storage of different lengths of slack of fiber optic cables such that a selected path length from the cable management structure to the curved portion of the cable support and back to the cable management structure substantially matches a length of slack of a fiber optic cable and such that the length of slack is maintained within the height of the base so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.

[0007] According to various aspects of the above embodiment, the standard rack length of the base is 19" or 23".

[0008] According to various aspects of any of the above embodiments, the cable support comprises a plurality of cable supports spaced apart along a length of the base.

[0009] According to various aspects of any of the above embodiments, each of the plurality of cable supports is configured to provide a different path length from the cable management structure to the curved portion of the cable support and back to the cable management structure so as to provide different path lengths that are configured to provide optimized management and storage of different lengths of slack of fiber optic cables

[0010] According to various aspects of any of the above embodiments, each of the cable supports comprises a D shape, and the curved portions of the cable supports face a same direction.

[0011] According to various aspects of any of the above embodiments, the panel further includes a second cable support comprising a D shape that is backward relative to the cable supports, and wherein the second cable support is disposed at the first end of the planar portion of the base between the cable management structure and one of the cable supports that is closest to the cable management structure.

[0012] According to various aspects of any of the above embodiments, the cable management structure comprises a clip configured to receive a fiber optic cable. In some aspects, the clip comprises two clips.

[0013] In accordance with various embodiments of the disclosure, a cable management panel for optimizing management and storage of a slack portion of a fiber optic cable includes a base that is configured to include a planar portion, a cable support that is configured to extend from the planar portion of the base, and a cable management structure that is disposed proximate an end of the planar portion and configured to extend from the planar portion of the base in a same direction as the cable support. The cable support includes a curved portion configured to prevent a fiber optic cable that is wrapped around the curved portion from being bent beyond a minimum bend radius. The cable support is configured to provide a path structure for optimizing management and storage of different lengths of slack of fiber optic cables such that a selected path length from the cable management structure to the curved portion of the cable support and back to the cable management structure substantially matches a length of slack of a fiber optic cable so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.

[0014] According to various aspects of any of the above embodiments, the panel further comprises a cable retaining structure that is disposed along a length of the planar portion in a longitudinal direction of the base and configured to extend from the planar portion of the base in the same direction as the cable support, and wherein the cable retaining structure is configured to retain a

slack fiber optic cable between the longitudinal edges of the base.

[0015] According to various aspects of any of the above embodiments, the base has a height equal to two rack units.

[0016] According to various aspects of any of the above embodiments, the base is configured to be mounted in a rack having a standard rack length of 19" or 23".

[0017] According to various aspects of any of the above embodiments, the cable support comprises a plurality of cable supports spaced apart along a length of the base.

[0018] According to various aspects of any of the above embodiments, each of the plurality of cable supports is configured to provide a different path length from the cable management structure to the curved portion of the cable support and back to the cable management structure so as to provide different path lengths that are configured to provide optimized management and storage of different lengths of slack of fiber optic cables.

[0019] According to various aspects of any of the above embodiments, each of the cable supports comprises a D shape, and the curved portions of the cable supports face a same direction.

[0020] According to various aspects of any of the above embodiments, the panel further comprises a second cable support comprising a D shape that is backward relative to the cable supports, and wherein the second cable support is disposed at the first end of the planar portion of the base between the cable management structure and one of the cable supports that is closest to the cable management structure.

[0021] According to various aspects of any of the above embodiments, the cable management structure comprises a clip configured to receive a fiber optic cable. In some aspects, the clip comprises two clips.

[0022] In accordance with various embodiments of the disclosure, a cable management panel for optimizing management and storage of a slack portion of a fiber optic cable includes a base portion and a cable support portion that is configured to extend from the base portion. The cable support includes a curved portion configured to prevent a fiber optic cable that is wrapped around the curved portion from being bent beyond a minimum bend radius. The cable support is configured to provide a path structure for optimizing management and storage of different lengths of slack of fiber optic cables such that a selected path length from an end of the base portion to the curved portion of the cable support and back to the end of the base portion substantially matches a length of slack of a fiber optic cable so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.

[0023] According to various aspects of any of the above embodiments, the panel further comprises a cable retaining structure that is disposed along a length of the base portion and configured to extend from the base portion in the same direction as the cable support, and wherein the cable retaining structure is configured to retain a slack fiber optic cable between longitudinal edges of the base portion.

[0024] According to various aspects of any of the above embodiments, the base portion has a height equal to two rack units.

[0025] According to various aspects of any of the above embodiments, the base portion is configured to be mounted in a rack having a standard rack length of 19" or 23".

[0026] According to various aspects of any of the above embodiments, the cable support comprises a plurality of cable supports spaced apart along a length of the base portion.

[0027] According to various aspects of any of the above embodiments, each of the plurality of cable supports is configured to provide a different path length from the end of the base portion to the curved portion of the respective cable support and back to the end of the base portion so as to provide different path lengths that are configured to provide optimized management and storage of different lengths of slack of fiber optic cables.

[0028] According to various aspects of any of the above embodiments, each of the cable supports comprises a D shape, and the curved portions of the cable supports face a same direction.

[0029] According to various aspects of any of the above embodiments, the panel further comprises a second cable support comprising a D shape that is backward relative to the cable supports, and wherein the second cable support is disposed between the end of the base portion and one of the cable supports that is closest to the end of the base portion.

[0030] According to various aspects of any of the above embodiments, the panel further comprises a cable management structure that is disposed proximate the end of the base portion and configured to extend from the base portion in a same direction as the cable support.

[0031] According to various aspects of any of the above embodiments, the cable management structure comprises a clip configured to receive a fiber optic cable. In some aspects, the clip comprises two clips.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] Further advantages and features of the present disclosure will become apparent from the following description and the accompanying drawings, to which reference is made. In which are shown:

[0033] FIG. 1 illustrates an exemplary optical fiber communication system including an exemplary cable management panel according to various aspects the present disclosure;

[0034] FIG. 2 is a front view of an exemplary cable management panel according to various aspects the present disclosure;

[0035] FIG. 3 is a bottom view of the cable management panel of FIG. 2;

[0036] FIG. 4 is a perspective view of the cable management panel of FIG. 2;

[0037] FIG. 5 is a perspective view of another exemplary cable management panel according to various aspects the present disclosure; and

[0038] FIG. 6 illustrates the exemplary cable management panel of FIG. 2 disposed in an optical fiber communication system in a lateral configuration.

DETAILED DESCRIPTION

[0039] Reference will now be made in detail to presently preferred embodiments and methods of the present disclosure, which constitute the best modes of practicing the present disclosure presently known to the inventors. It is to be understood that the disclosed embodiments are merely exemplary of the present disclosure that may be embodied in various and alternative forms.

Therefore, specific details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for any aspect of the present disclosure and/or as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

[0040] It is also to be understood that this present disclosure is not limited to the specific embodiments and methods described below, as specific components and/or conditions may, of course, vary. Furthermore, the terminology used herein is used only for the purpose of describing particular embodiments of the present disclosure and is not intended to be limiting in any way. It must also be noted that, as used in the specification and the appended claims, the singular form “a,” “an,” and “the” comprise plural referents unless the context clearly indicates otherwise. For example, reference to a component in the singular is intended to comprise a plurality of components.

[0041] FIG. 1 illustrates an exemplary optical fiber communication system **100**. The system includes one or more racks **110** of optical fiber connectivity modules **112** and passageways, raceways, or ducting **114** around the racks **110** through which cables, for example, optical fiber cables, may be routed. Depending on the size of the enclosure containing the rack(s) **110** and/or the amount of space available around the rack(s) **110**, for example, above, below, left, and right of the rack(s) **110**, conventional slack fiber cable management structures, such as spools, may not be

usable in such space around the rack(s) **110**.

[0042] As illustrated in FIG. 1, a cable management panel **116** in accordance with the present disclosure is configured to be mounted to a rack **110**, for example, in an active street cabinet or an equipment rack with low to medium fiber density. The cable management panel **116** may be configured as a 2 rack unit (or 2U) panel, for example, a panel having a 2U height and a standard rack mountable length, for example, 19" or 23", where one rack unit is equal to 1.75 inches. In some aspects, the space available around the rack **110** for fiber cable slack management may be insufficient for conventional slack fiber cable management structures, such as spools.

[0043] The panel **116** is configured to store and/or manage slack cable (or a slack portion of cable), for example, fiber optic cables such as patch cables, pigtails, and the like. The fiber optic cables may be preconnectorized and/or ruggedized, as would be understood by persons skilled in the art.

[0044] Referring now to FIGS. 2-4, the cable management panel **116** may include a base or base portion **117**, a first cable support or first cable support portion **118**, and a second cable support or second cable support portion **120**. In the illustrated embodiment, the cable management panel **116** includes four first cable supports **118** and one second cable support **120**. Of course, it should be understood that the panel **116** may include more or less than four first cable supports **118** and/or more than one second cable support **120**. The base **117** may include a planar portion **126** and a curved portion **127** at a first end **160** of the planar portion **126**. The planar portion **126** may be sized and arranged with a radius of curvature configured such that an optical fiber cable being wrapped around the curved portion **127** is prevented from being bent beyond a normal operation minimum bend radius. As would be appreciated by persons skilled in the art, if an optical fiber cable **18** is bent beyond the minimum bend radius, signal degradation, signal loss, and/or a broken fiber may result.

[0045] The first cable supports **118** and the second cable support **120** may extend substantially perpendicularly from a first side **119** of the base **117**, for example, the planar portion **126** of the base **117**, as best shown in FIG. 3. The first cable supports **118** are spaced apart sequentially along a longitudinal direction of the base **117** from a second end **162** of the planar portion **126** toward the first end **160**. The second cable support **120** is spaced from the first cable support **118'** that is nearest the first end **160** and is between the first cable support **118'** and the curved portion **127**.

[0046] As illustrated in FIG. 2, the first cable supports **118** are configured in a backward D shape when the panel **116** is viewed with the curved portion **127** to the right of the planar portion **126**. The second cable support **118** is configured in a D shape when the panel **116** is viewed with the curved portion **127** to the right of the planar portion **126**. For example, the first cable supports **118** include a curved portion **164** and a planar portion **166** that form the D shape, and the second cable support **120** includes a curved portion **168** and a planar portion **170** that form the D shape. The curved portions **164**, **168** are sized and arranged with a radius of curvature configured such that an optical fiber cable **18** being wrapped around any of the curved portions **164**, **168** is prevented from being bent beyond a normal operation minimum bend radius. As would be appreciated by persons skilled in the art, if an optical fiber cable **18** is bent beyond the minimum bend radius, signal degradation, signal loss, and/or a broken fiber may result.

[0047] The first and second cable supports **118**, **120** may include a retaining feature **140** at their end surfaces furthest from the first side **119** of the base **117**. The retaining feature **140** may comprise a tab extending outward from the first and/or second cable supports **118**, **120**, for example, the curved portions **164**, **168**, and substantially parallel to the planar portion **170** of the base **117** such that the retaining feature **140** is configured to retain a fiber optic cable between the retaining feature **140** and the first side **119** of the base **117**. In some embodiments, the retaining features **140** may include more than one tab extending from the curved portion **164** or **168**.

[0048] As shown, the planar portion **170** of the second cable support **120** faces in a direction toward the first cable supports **118**, and the curved portion **168** of the second cable support **120** faces in a direction away from the first cable supports **118**. Similarly, the planar portions **166** of the

first cable supports **118** face in a direction toward the second cable support **120**, and the curved portions **164** of the first cable supports **118** face in a direction away from the second cable support **120**. The first cable supports **118** and the second cable support **120** are configured to provide multiple path lengths for managing and/or storing various slack lengths of optical fiber cable **18**, as will be described in more detail below.

[0049] The panel **116** may include a retaining structure **122** along a longitudinal edge **124** of the panel **116**, for example, the planar portion **170** of the panel **116**. The retaining structure **122** is configured to retain a portion of the slack length of optical fiber cable **118** in the 2U height of the panel **116** for management and storage. In some embodiments, the retaining structure **122** may comprise a substantially planar tab **123** extending substantially perpendicularly from the first side **119** of the base **117**, for example, the planar portion **170** of the base **117**, in the same direction as the first and second cable supports **118**, **120**. The free end of the retaining structure **122** may comprise a lip **125** facing inward relative to the longitudinal edge **124**. The retaining structure **122** may comprise a plurality of spaced apart retaining structures **122**, as illustrated in FIGS. 2-4, or a single elongated retaining structure **122'**, as shown in FIG. 5. In some embodiments, the panel **116** may include a retaining structure **122** along both longitudinal edges **124**, as shown in FIGS. 2-4, while in other embodiments, a panel **116'** may include a retaining structure **122** along only one of the longitudinal edges **124**, as shown in FIG. 5.

[0050] The panel **116** may include a cable management structure **136** at the first side **119** of the base **117**, for example, the planar portion **170** of the base **117** between the second cable support **120** and the first end **160** of the planar portion **170**. In some embodiments, the cable management structure **136** may comprise a clip or bracket **138** extending substantially perpendicularly from the first side **119** of the base **117**, for example, the planar portion **170** of the base **117**, in the same direction as the first and second cable supports **118**, **120** and the retaining structure **122**. The clip **138** may include an opening configured to receive a fiber optic cable and, in some embodiments, the clip **138** may be configured such that the opening is closable. In some embodiments, the cable management structure **136** may include two clips **138** that are spaced apart from one another in a direction from one of the longitudinal edges **124** to the other, as best illustrated in FIG. 2. The cable management structure **136** may be configured such a first portion of a fiber optic cable **18** can pass through one clip **138**, a second portion can wrap around a first cable support **118**, and a third portion can pass through the other clip, as will be described in more detail below.

[0051] In use, as illustrated in FIG. 1, the panel **116** is configured to be mounted horizontally in a rack **110** of an optical fiber communication system **100**. The panel **116** may be mounted in a first horizontal orientation **180** where the second cable support **120** and the first end **160** of the planar portion **170** are at the right side of the rack **110** or in a second horizontal orientation **182** where the second cable support **120** and the first end **160** of the planar portion **170** are at the left side of the rack **110**.

[0052] As shown in FIG. 1, the panel **116** is configured to manage various different lengths of slack cable from various different lengths of preconnectorized fiber optic cables **18**, for example, patch cables or pigtails. For example, two fiber optic cables **18a** having the same length are received by panels **116** in the first horizontal orientation **180**. The fiber optic cables **18a** are fed from a first connectivity module **112a** through a first clip **138** of the cable management structure **136**, wrapped around the left-most one of the first cable supports **118**, and fed back through a second clip **138** of the cable management structure **136** to a second connectivity module **112b**.

[0053] Another fiber optic cable **18b** having a length that is shorter than that of the fiber optic cable **18a** similarly is fed from a first connectivity module **112c** through a first clip **138** of the cable management structure **136**, is wrapped around the third from the right of the first cable supports **118**, and is then fed back through a second clip **138** of the cable management structure **136** to a second connectivity module **112d**.

[0054] In another arrangement, two fiber optic cables **18a** having different lengths are received by

panels **116** in the first horizontal orientation **180**. A fiber optic cable **18c** having a first length of slack cable is fed from a first connectivity module **112e** through a first clip **138** of the cable management structure **136**, wrapped around the right-most one of the first cable supports **118**, and fed back through a second clip **138** of the cable management structure **136** to a second connectivity module **112f**. Another fiber optic cable **18d** having a length of slack cable that is longer than that of the fiber optic cable **18c** similarly is fed from the first connectivity module **112e** through a first clip **138** of the cable management structure **136**, is wrapped around the second from the right of the first cable supports **118**, and is then fed back through a second clip **138** of the cable management structure **136** to the second connectivity module **112f**.

[0055] As illustrated, the cables **18a**, **18b**, **18c**, **18d** are selectively wrapped around the one of the first cable supports **118** such that management and storage of the slack of the cables **18a**, **18b**, **18c**, **18d** is optimized; that is, a path length from the cable management structure **136** to the curved portion **164** of the selected one of the first cable supports **118** (such that the slack can be wrapped around the curved portion **164**) and back to the cable management structure **136** substantially matches a length of the slack of the cable **18a**, **18b**, **18c**, **18d**. In such an optimized management and storage arrangement, the slack of the cables **18a**, **18b**, **18c** may have a minimized amount of droop such that the slack remains between the longitudinal edges **124** of the base **117**. In some aspects, the retaining structure **122** may assist with maintaining the slack of the cables **18a** between the longitudinal edges **124**.

[0056] In some aspects, the management and storage of the slack of the cables **18a**, **18b**, **18c**, **18d** is optimized when the slack of the cables **18a**, **18b**, **18c**, **18d** takes the shortest path from the cable management structure **136** to being wrapped around a curved portion **164** of a first cable support **118** and back to the cable management structure **136** so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted. In some aspects, the management and storage of the slack of the cables **18a**, **18b**, **18c** is optimized when the slack of the cables **18a**, **18b**, **18c** has the least amount of droop as the slack extends from the cable management structure **136** to being wrapped around a first cable support **118** (and optionally the second cable support) and back to the cable management structure **136** so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.

[0057] Although not illustrated, it should be appreciated that a panel **116** can receive a plurality of slack lengths of fiber optic cables **18**, and the fiber optic cables may have the same or different lengths and/or the slack portions of the fiber optic cables may have the same or different lengths. Further, it should be understood that a slack portion of a fiber optic cable may be sufficiently long such that when the cable is fed through the cable management structure **136** and wrapped around a furthest one of the first supports **118**, a remaining length of the slack portion may still need to be managed before the cable is fed back through the cable management structure **136**. In such a situation, the remaining length of the slack portion may be wrapped around the second support **120** and then wrapped around one of the first supports **118** before being fed through the cable management structure **136**.

[0058] In the above embodiments, the panel **116** is configured to optimize management and storage of the slack of a fiber optic cable by providing a path length from the cable management structure **136** to the curved portion **164** of a first cable support **118** (such that the slack can be wrapped around the curved portion **164**) and back to the cable management structure **136** that substantially matches a length of the slack of the cable **18** so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.

[0059] As described, the panel **116** may be reversibly mounted to a rack **110** such that the cable management structure **136** may be at the right side or the left side of the rack **110**, thereby permitting fibers to be routed to the panel **116** from the right side or the left side of the rack **110**.

[0060] Referring now to FIG. **6**, the panel **116** may also be arranged in a vertical orientation **184** to provide management and/or storage of a cable **18** laterally of the rack **110**. As a result, the rack **110**

includes more space for mounting connectivity modules **112**. In the vertical orientation, the panel **116** permits various entry points for the cable, and the first and second supports **118**, **120** permit various lengths of slack cable to be wrapped around the curved portion **164**, **168** of a first cable support **118** and/or second cable support **120**.

[0061] It should be appreciated that in some aspects, the panel **116** may be a single, monolithic piece of unitary construction such that the base **117**, the first supports **118**, the second supports **120**, the retaining structure **122**, and the cable management structure **136** comprise a single, monolithic piece of unitary construction. In some aspects, the base **117**, the first supports **118**, the second supports **120**, the retaining structure **122**, and/or the cable management structure **136** may be constructed as separate elements and assembled into an integral structure via any conventional attachment means, for example, a screw, a snap connection, or the like.

[0062] It should be appreciated that the first and second support elements **118**, **120** may be hollow structures, as shown, or may be solid structures in other non-illustrated embodiments. In some aspects, the planar portions **166**, **170** may be omitted such that the first and second support elements **118**, **120** include only the curved portions **164**, **168**.

[0063] While at least one example, non-limiting embodiment has been presented in the foregoing detailed description, it should be appreciated that a vast number of variations exist. It should also be appreciated that the exemplary embodiment or exemplary embodiments are only examples, and are not intended to limit the scope, applicability, or configuration of the disclosure in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing the exemplary embodiment or exemplary embodiments. It should be understood that various changes can be made in the function and arrangement of elements without departing from the scope of the disclosure as set forth in the appended claims and the legal equivalents thereof.

Claims

1. A cable management panel for optimizing management and storage of a slack portion of a fiber optic cable, comprising: a base that is configured to include a planar portion having a first end and a second end in a longitudinal direction and a curved portion extending from the first end of the planar portion; a cable support that is configured to extend substantially perpendicularly from the planar portion of the base; a cable management structure that is disposed proximate the first end of the planar portion and configured to extend substantially perpendicularly from the planar portion of the base in a same direction as the cable support; a cable retaining structure that is disposed along a longitudinal edge of the planar portion and configured to extend substantially perpendicularly from the planar portion of the base in the same direction as the cable support; wherein the base has a height equal to two rack units; wherein the base is configured to be mounted in a rack having a standard rack length; wherein the cable retaining structure is configured to retain a slack fiber optic cable between the longitudinal edge and an opposed longitudinal edge of the base; wherein the curved portion of the base is configured to prevent a fiber optic cable that is wrapped around the curved portion of the base from being bent beyond a minimum bend radius; wherein the cable support includes a curved portion configured to prevent a fiber optic cable that is wrapped around the curved portion from being bent beyond a minimum bend radius; and wherein the cable support is configured to provide a path structure for optimizing management and storage of different lengths of slack of fiber optic cables such that a selected path length from the cable management structure to the curved portion of the cable support and back to the cable management structure substantially matches a length of slack of a fiber optic cable such that the length of slack is maintained within the height of the base so as to eliminate the need for slack cable management structures in space about a rack to which the panel is mounted.
